# Signals and Systems
# SET214
# Lecture (05)- Properties of Signals


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# Outlines

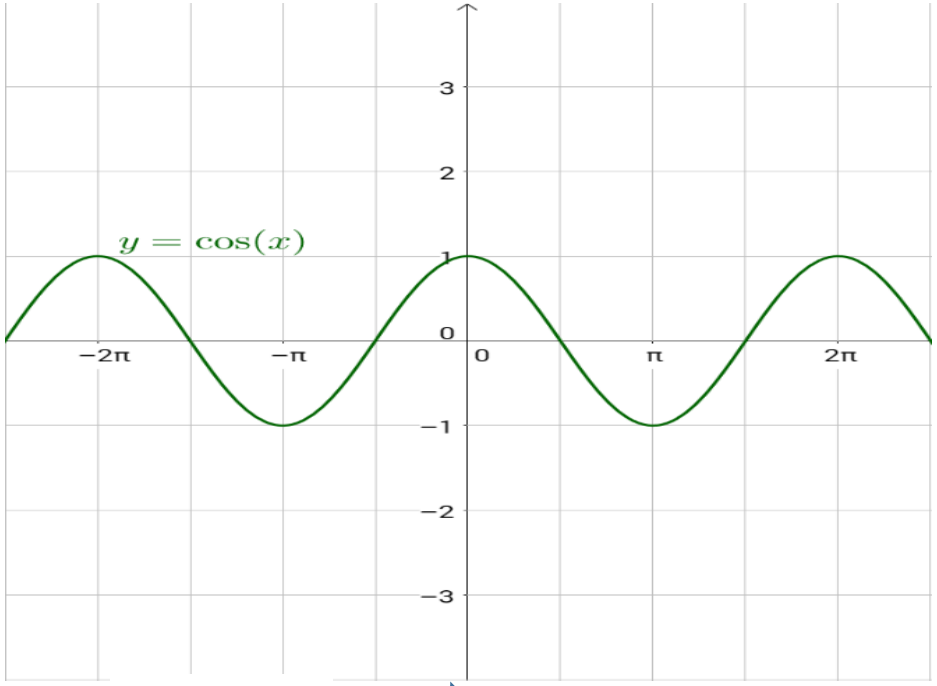
- Even Symmetry
- Odd Symmetry
- Exercise
- Even and Odd Properties
- Periodic Signal
- Sum of Periodic Signals
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- Energy and Power Signals
- Exercise

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# Even Symmetry

- A function *x* is said to be even if it satisfies. $x(t) = x(-t)$ for all *t* (where *t* is a real number)
- Geometrically, the graph of an even signal is symmetric with respect to the vertical axis.
  - Ex, $x(t) = \cos(w_0 t)$ → even signal

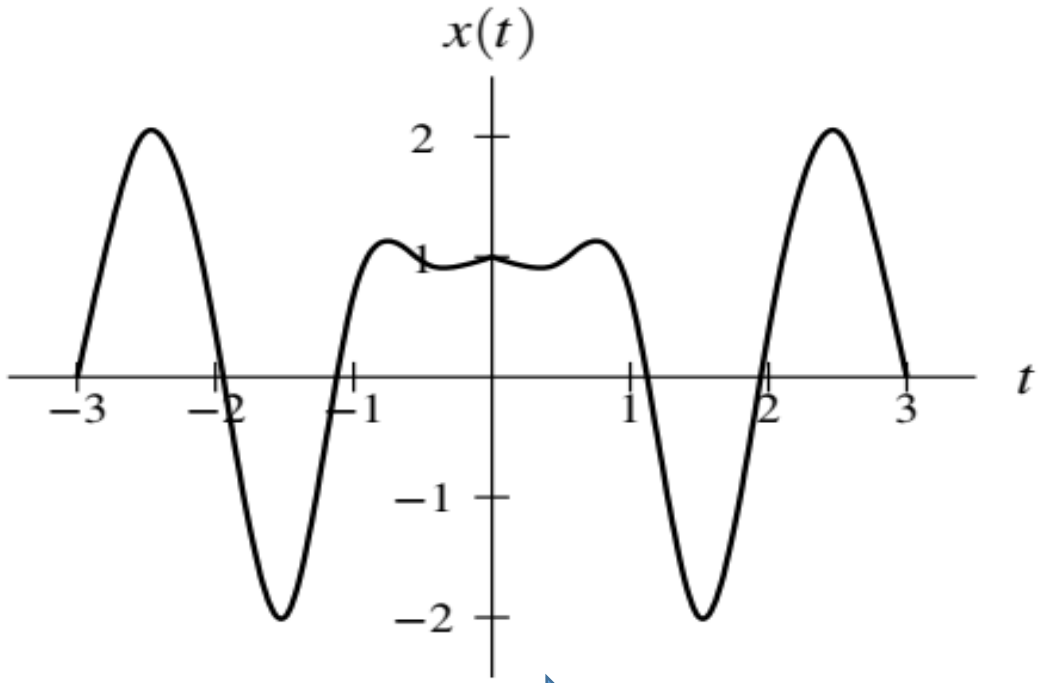



X[π] =-1

X[- π]=-1

even

*symmetric* with respect to the vertical axis.




X[2] =0

X[-2]=0

even

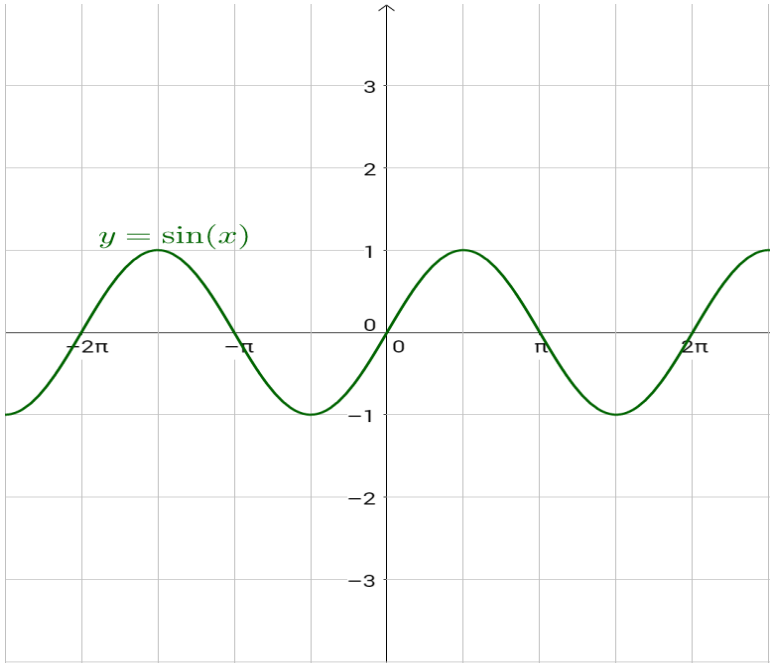
*symmetric* with respect to the vertical axis.

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# Odd Symmetry

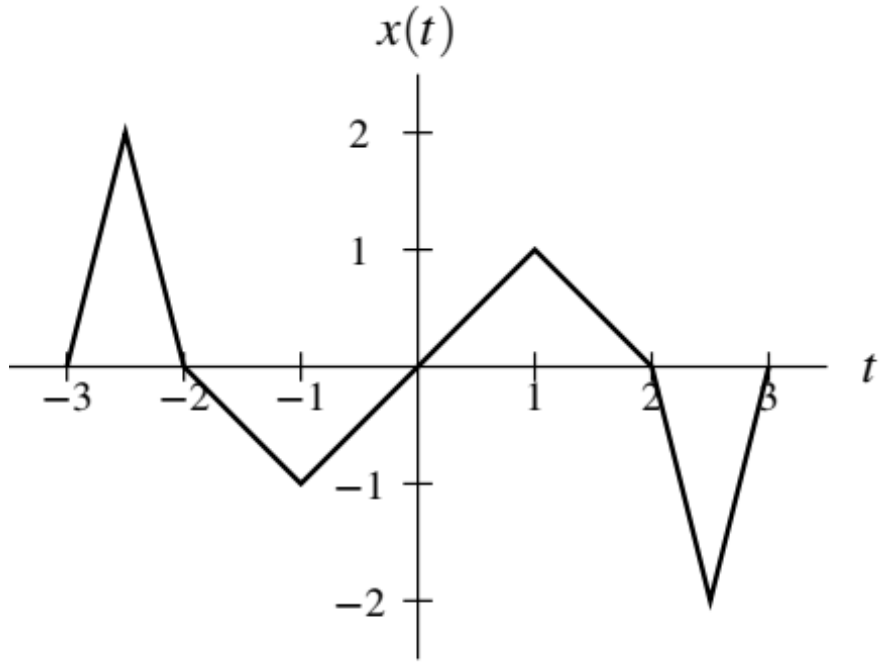
- A function *x* is said to be odd if it satisfies $x(t) = -x(-t)$ for all *t* (where *t* is a real number)
- An odd signal *x* must be such that $x(0) = 0$
- Geometrically, the graph of an odd signal is symmetric with respect to the origin.
  - Ex, x(t)= sin ($\omega_o$t)→ odd signal




X[$\pi/2$] =1
X[$-\pi/2$] = -1

→ X[$\pi/2$]= - X[$-\pi/2$] **odd**

*symmetric* with respect to the origin




X[1] =1
X[-1] = -1

→ X[1]= - X[$-1$] **odd**

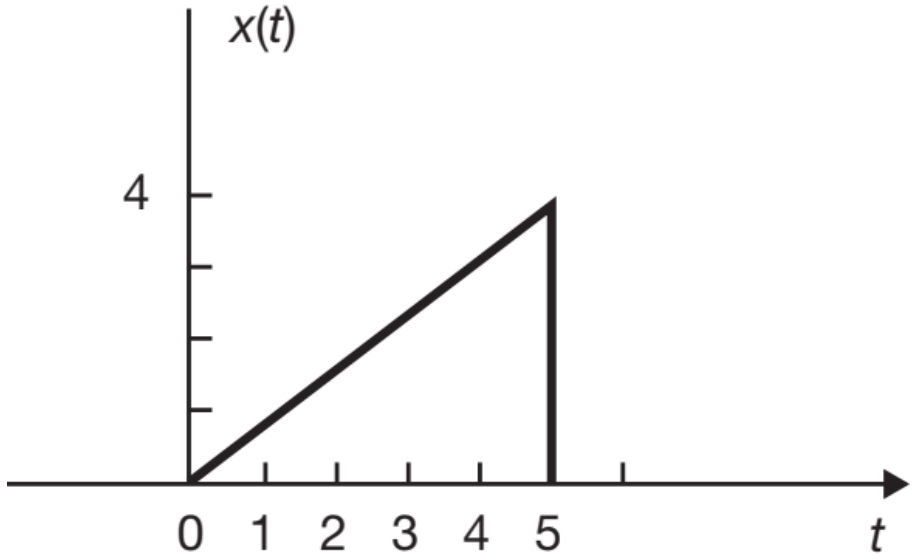
*symmetric* with respect to the origin

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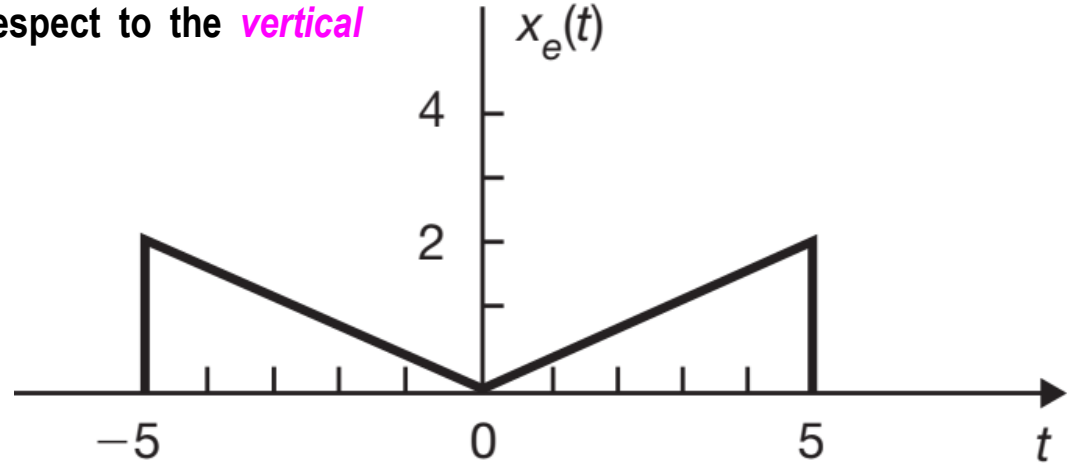
# Exercise (1/7)

- Sketch and label the even and odd components of the signals shown in Fig.

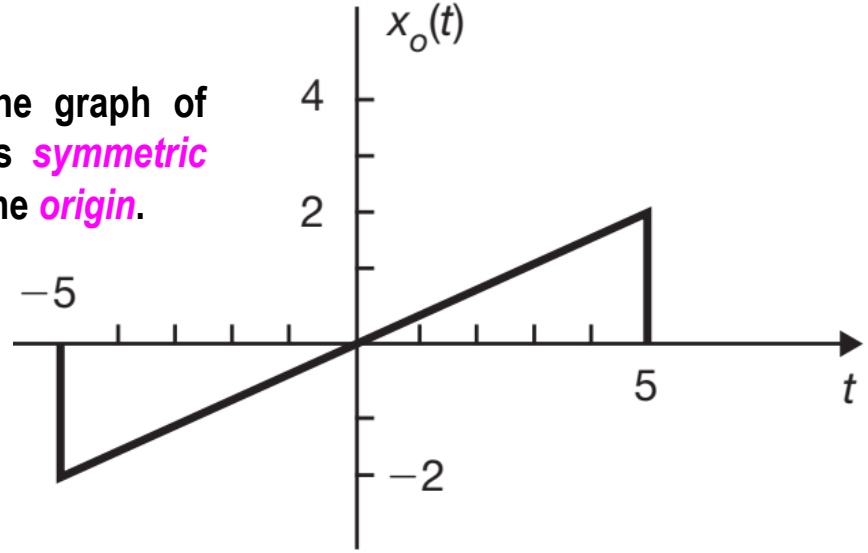



**Ans.**

- Geometrically, the graph of an even signal is *symmetric* with respect to the *vertical axis.*

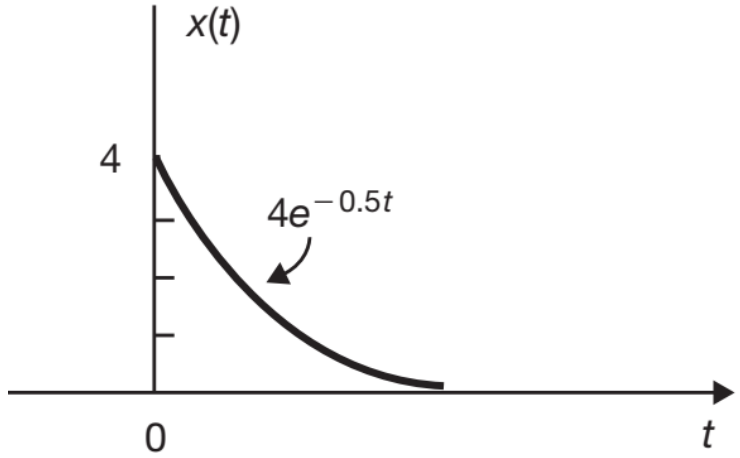



- Geometrically, the graph of an odd signal is *symmetric* with respect to the *origin*.



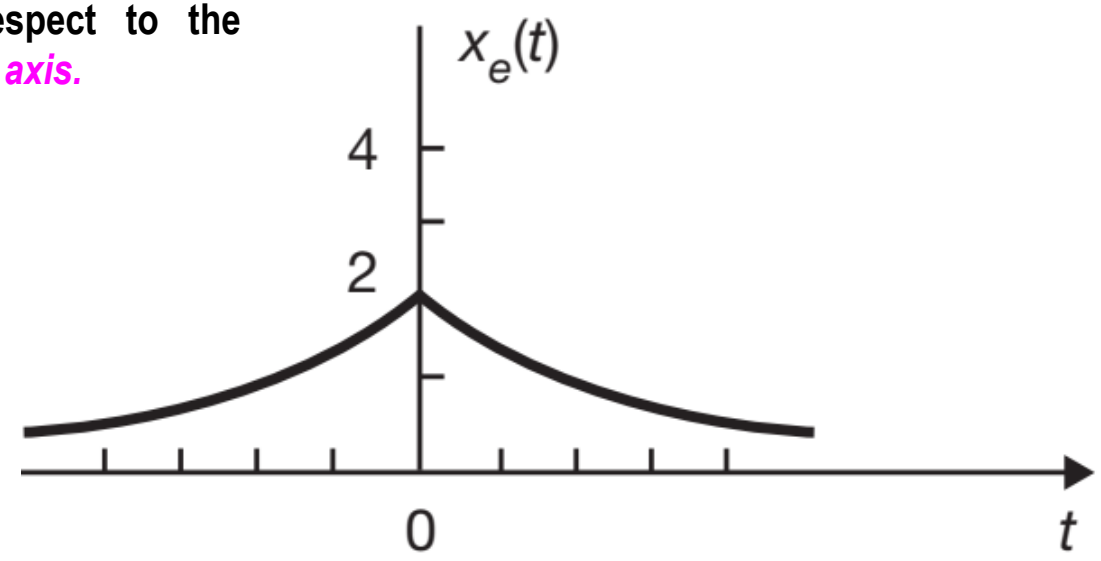
# Exercise (2/7)

- Sketch and label the even and odd components of the signals shown in Fig.

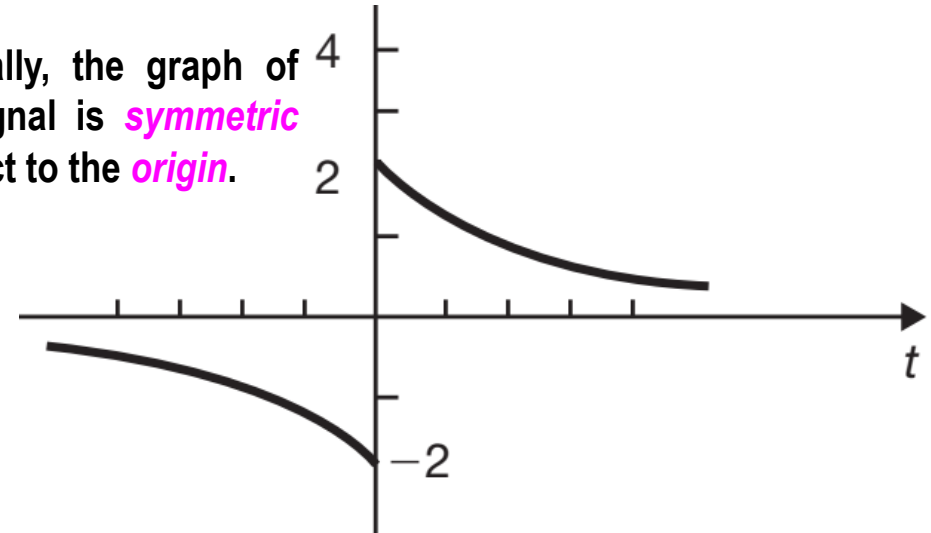



**Ans.**

- Geometrically, the graph of an even signal is *symmetric* with respect to the *vertical axis.*

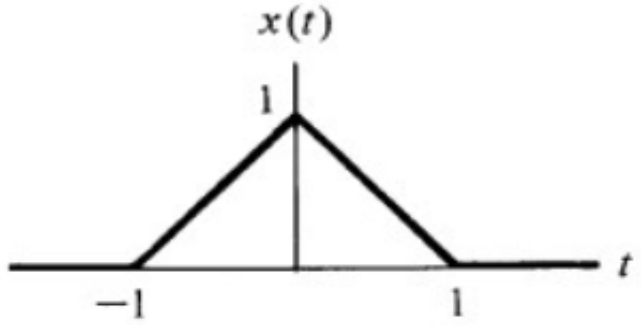



- Geometrically, the graph of an odd signal is *symmetric* with respect to the *origin*.

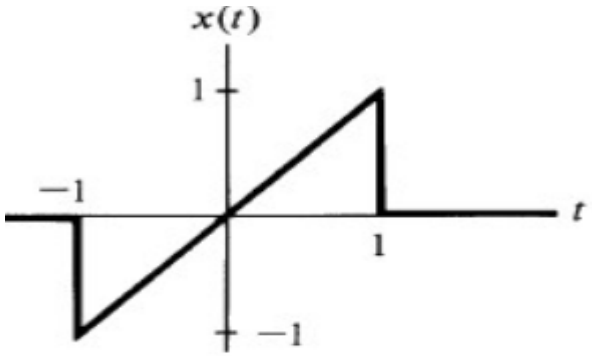


# Exercise (3/7)

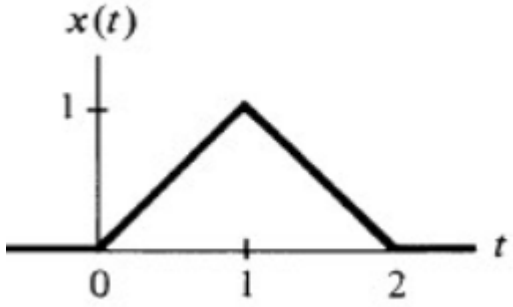
- For each of the following signals, determine whether it is even, odd, or neither




(A)




(B)




(C)

**Ans.**

By definition a signal is even if and only if x(t)=x(-t), while a signal is odd if and only if x(t)=-x(-t)

(A)Since x(t) is symmetric around vertical axis, x(t)=x(-t)→ x(t) is even
(B)Since x(t) is symmetric around origin, x(t)=-x(-t)→ x(t) is odd
(C)It is reality seen that x(t)≠ x(-t) for all t, and x(t) ≠ - x(-t) for all t → x(t) is neither even nor odd

# Exercise (4/7)

- Check whether the given signal is even, odd, or neither

(a) X(t)=2

Ans.

X(t)=2

X(-t)= 2

X(t)=x(-t) → signal is even

(b) $X(t)=\sin(t^2)$

Ans.

$X(t)=\sin(t^2)$

$X(-t)= \sin((-t)^2) =\sin(t^2)$

X(t)=x(-t) → signal is even

# Exercise (5/7)

- Check whether the given signal is even, odd, or neither

(c) $X(t)=2t^3$

**Ans.**

$X(t)=2t^3$

$X(-t)= 2(-t)^3 = -2t^3$

$X(t)=- x(-t)$ → signal is odd

(d) $X(t)=2t^2$

**Ans.**

$X(t) =2t^2$

$X(-t)= 2(-t)^2 = 2t^2$

$X(t) = x(-t)$ → signal is even

# Exercise (6/7)

- Check whether the given signal is even, odd, or neither

(e) X(t)=5 sin(wt)

Ans.

X(t)=5 sin(wt)
X(-t)= 5 sin(-wt)= - 5 sin(wt)
X(t)= - x(-t) → signal is odd

(f) X(t)=3cos (wt)

Ans.

X(t)=3cos (wt)
X(-t)= 3cos (-wt)=3cos (wt)
X(t)= x(-t) → signal is even

# Exercise (7/7)

- Check whether the given signal is even, odd, or neither

## (g) $X(t)=\sin^2(t)$

**Ans.**

$\sin^2(t)=\frac{1}{2}(1-\cos(2t))$

$X(t)=\frac{1}{2}(1-\cos(2t))$

$X(-t)=\frac{1}{2}(1-\cos(2(-t)))$

$=\frac{1}{2}(1-\cos(2t))$

$X(t)= x(-t)$ → signal is even

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# Even and Odd (1/2)

- **Sums** involving even and odd functions have the following properties:
  - The sum of two even functions is even . Ex, $\cos(wt)+ t^2 \rightarrow$ even
  - The sum of two odd functions is odd . Ex, $\sin(wt)+ t^3 \rightarrow$ odd
  - The sum of an even function and odd function is neither even nor odd, provided that neither of the functions is identically zero.

- **Products** involving even and odd functions have the following properties:
  - The product of two even functions is even . Ex, $t^2 * t^4 = t^6 \rightarrow$ even
  - The product of two odd functions is even . Ex, $t^3 * t^5 = t^8 \rightarrow$ even
  - The product of an even function and an odd function is odd . Ex, $t^2 * t^3 = t^5 \rightarrow$ odd

# Even and Odd Prosperities (1/2)

- Every function x has a unique representation of the form $x(t) = x_e(t) + x_o(t)$;
- where the functions $x_e$ and $x_o$ are even part and odd part , respectively.
- In particular, the functions $x_e$ and $x_o$ are given by:-

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

- Find the even and odd part of the following signal:-

$$X(t) = 3 - \frac{t^2}{\sin(t)} + \frac{\cos(t)}{t} - \frac{\sin^2(t)}{t^4} + t^3.\sin^3(t)$$

**Ans.**

$$X(t) = 3 - \frac{t^2}{\sin(t)} + \frac{\cos(t)}{t} - \frac{\sin^2(t)}{t^4} + t^3.\sin^3(t)$$

| | | | | |
|---|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ | ↓ |
| E | $\frac{E}{O}$ | $\frac{E}{O}$ | $\frac{E}{E}$ | $O * O$ |
| ↓ | ↓ | ↓ | ↓ | ↓ |
| E | 0 | 0 | $E$ | E |

$$Xe(t) = 3 - \frac{\sin^2(t)}{t^4} + t^3 * \sin^3(t)$$

$$Xo(t) = -\frac{t^2}{\sin(t)} + \frac{\cos(t)}{t}$$

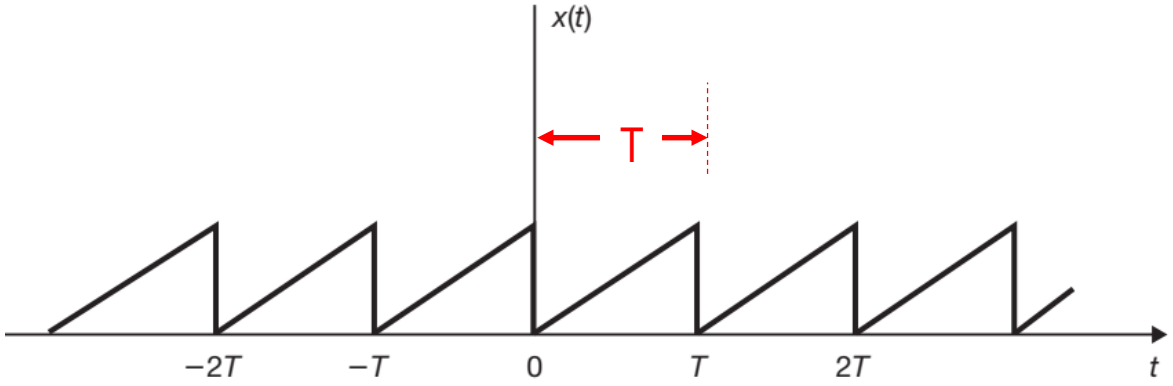
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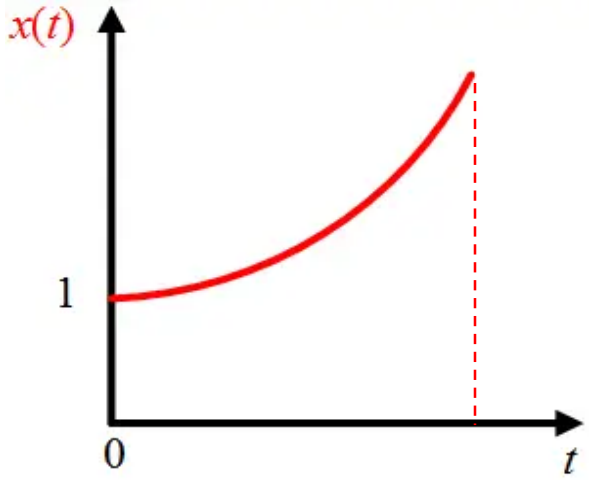
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# Periodic Signal (1/2)

- A function *x* is said to be periodic with period *T* (or *T*-periodic) if the following condition holds,

$$x(t) = x(t + T) \text{ for all t (where t is a real number)}$$



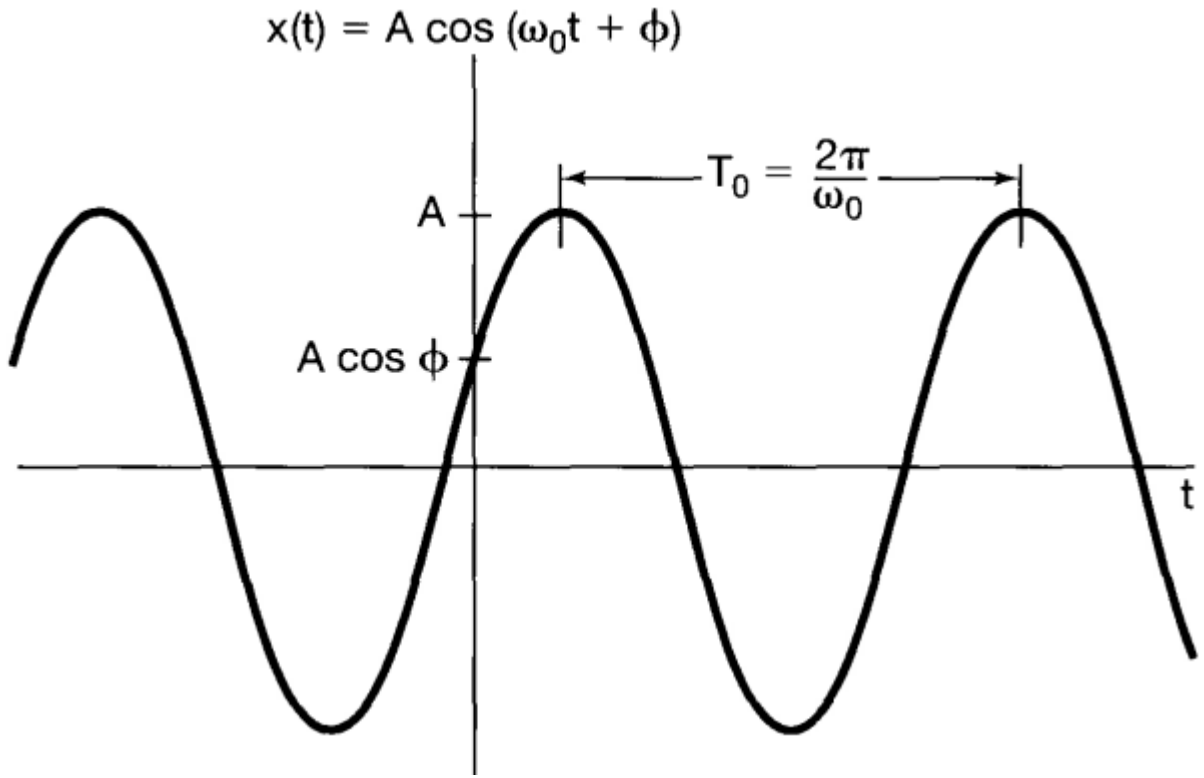




- A *T*-periodic function *x* is said to have frequency $1/T$ and angular frequency $2\pi/T$.
- A function/sequence that is not periodic is said to be aperiodic.

# Periodic Signal (2/2)

- A: amplitude
- $\omega_o = 2\pi f o$: Angular_frequency
- ∅: Phase shift
- $T_o=1/f_o$ : Time period

- Fundamental Time Period (FTP): It is the smallest positive and fixed value of the time for which signal is periodic




$$\omega o=2\pi f o = \frac{2\pi}{To}$$

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# Exercise (1/3)

❑ Find FTP of the following signal

(A) X(t)=A sin(2$\pi$t)

Ans.

X(t)=A sin( $\omega_o$t +∅)

$\omega o=2\pi$ ⇒ $\omega o=2\pi f o=\frac{2\pi}{To}$

$To=\frac{2\pi}{wo}=\frac{2\pi}{2\pi}=1$ sec

(b) X(t)=A cos(10$\pi$t+30)

Ans.

X(t)=A cos( $\omega_o$t +∅)

$\omega o=10\pi$ ⇒ $\omega o=2\pi f o=\frac{2\pi}{To}$

$To=\frac{2\pi}{10\pi}=0.2$ Sec

# Exercise (2/3)

❑ Find FTP of the following signal

(c) X(t)= -30 cos(0.2$\pi$t+30)

Ans.

X(t)= A cos( $\omega_o$t +∅)

$\omega o$=0.2$\pi$ ⇒ $\omega o = 2\pi f_o = \frac{2\pi}{To}$

$To = \frac{2\pi}{0.2\pi}$=10 Sec

(d) X(t)= 6+ 2 cos(0.4$\pi$t+15)

Ans.

X(t)= Dc + A cos( $\omega_o$t +∅)

$\omega o$=0.4$\pi$ ⇒ $\omega o = 2\pi f_o = \frac{2\pi}{To}$

$To = \frac{2\pi}{0.4\pi}$=5 Sec

# Exercise (3/3)

❑ Find FTP of the following signal

(e) X(t)= $\sin^2(4\pi t)$

**Ans.**

$\sin^2(wt) = 0.5(1 - cos(2wt)$

$\sin^2(4\pi t) = 0.5(1 - cos(8\pi t)$

X(t)= $0.5(1 - cos(8\pi t)$

X(t)= $0.5 - 0.5\ cos(8\pi t)$

X(t)= $Dc - A\ cos(w_o t + \emptyset)$

$\omega o = 8\pi$ ⇒ $\omega o = 2\pi f_o = \frac{2\pi}{To}$

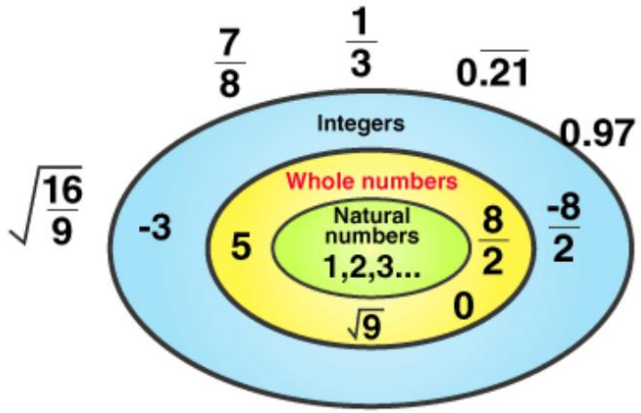
$To = \frac{2\pi}{wo} = \frac{2\pi}{8\pi} = 0.25$ sec

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# Sum of Periodic Signals

- For two periodic continuous functions x1 and x2 with fundamental periods T1 and T2, respectively, the sum y = x1 +x2.
- The sum y is periodic *if and only if the ratio* T1=T2 is a rational number
  - Note→ e,π, $\sqrt{2}$, $\sqrt{1.6}$ ,….etc are irrational number
- Suppose now that y is periodic. Let $\frac{T1}{T2} = \frac{q}{r}$ where q and r are integers and coprime →Then, x is periodic with period T = rT1 = qT2.




rational number

$\sqrt{8}$

$\sqrt{1.6}$ $\sqrt{11}$

$\sqrt{\frac{2}{5}}$ π

0.3030030003

irrational number

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# Exercise (1/3)

Let x1(t) = sin( π t) and x2(t) = sin(t). Determine whether the function y = x1 +x2 is periodic. Also, if it is, find its fundamental period.

**Ans.**

Denote the fundamental periods of x1 and x2 as T1 and T2, respectively. We then have

$$T_o = \frac{2\pi}{w_o} \Rightarrow T_1 = \frac{2\pi}{\pi} = 2$$

$$\Rightarrow T_2 = \frac{2\pi}{1} = 2\pi$$

$$\frac{T_1}{T_2} = \frac{r}{q} \Rightarrow \frac{T_1}{T_2} = \frac{2}{2\pi} = \frac{1}{\pi}$$

Since $\pi$ is an irrational number, $\frac{T_1}{T_2}$ is not rational. Therefor y is not periodic

# Exercise (2/3)

Let $x1(t) = \cos(2\pi t + 4)$ and $x2(t) = \sin(7\pi t)$. Determine if the function $y = x1 + x2$ is periodic, and if it is, find its fundamental period.

**Ans.**

Denote the fundamental periods of $x1$ and $x2$ as $T1$ and $T2$, respectively. We then have

$$T_o = \frac{2\pi}{w_o} \Rightarrow T_1 = \frac{2\pi}{2\pi} = 1$$

$$\Rightarrow T_2 = \frac{2\pi}{7\pi} = \frac{2}{7}$$

$$\frac{T_1}{T_2} = \frac{r}{q} \Rightarrow \frac{T_1}{T_2} = \frac{7}{2}$$

$\frac{T_1}{T_2}$ is rational. Thus, y is periodic

$$\therefore T_o = 2T_1 = 7T_2 = 2$$

# Exercise (3/3)

Let $x1(t) = \cos(6\pi t)$ and $x2(t) = \sin(30\pi t)$. Determine if the function $y = x1 + x2$ is periodic, and if it is, find its fundamental period.

**Ans.**

Denote the fundamental periods of $x1$ and $x2$ as $T1$ and $T2$, respectively. We then have

$$T_o = \frac{2\pi}{w_o} \Rightarrow T_1 = \frac{2\pi}{6\pi} = \frac{1}{3}$$

$$\Rightarrow T_2 = \frac{2\pi}{30\pi} = \frac{1}{15}$$

$$\frac{T_1}{T_2} = \frac{r}{q} \Rightarrow \frac{T_1}{T_2} = \frac{15}{3} = \frac{5}{1}$$

$\frac{T_1}{T_2}$ is rational. Thus, y is periodic

$$T_o = 1T_1 = 5T_2 = \frac{1}{3}$$

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# Energy and Power Signals (1/2)

- Consider $v(t)$ to be the voltage across a resistor $R$ producing a current $i(t)$. The instantaneous power $p(t)$ per ohm is defined as:-

$$p(t) = \frac{v(t)i(t)}{R} = i^2(t) = |\mathbf{X(t)}|^2$$

The **energy** $E$ contained in the function $x$ is given by :-

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt.$$

**A signal with finite energy is said to be an energy signal**

**The average power P contained in the function x is given by:-**

$$P = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt.$$

**Non-periodic**

**A signal with (nonzero) finite average power is said to be a power signal**

$$P = \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt.$$

**Periodic**

# Energy and Power Signals (2/2)

- **Energy Signal**
  - A signal is said to be an energy signal if and only if its total energy E is finite, i.e., $0 < E < \infty$.
  - For an energy signal, the average power P = 0.
  - The nonperiodic signals are the examples of energy signals

- **Power Signal**
  - A signal is said to be a power signal if its average power P is finite, i.e., $0 < P < \infty$.
  - For a power signal, the total energy $E = \infty$.
  - The periodic signals are the examples of power signals.

- **Noted:**
  - The signal for which both power and energy tends to infinite is neither energy signal nor power signal i.e. $E = \infty$ and $P = \infty$
  - Energy signal and power signals are mutually exclusive.
  - For some signals energy is finite and for some signals power is finite but no signal posses the finite power and finite energy.

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# Exercise (1/2)

❑ Determine whether the following signals are energy signals, power signals, or neither.

$$x(t) = e^{-at}u(t), \quad a > 0$$

**Ans.**

X(t) is not periodic due to u(t)

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |e^{-at}\, u(t)\,|^2 dt$$

$$E = \int_{0}^{\infty} |e^{-at}|^2 dt = \int_{0}^{\infty} e^{-2at} dt =$$

$$E = \int_{0}^{\infty} e^{-2at} dt = \frac{1}{-2a} * e^{-2at}\Big|_{0}^{\infty}$$

$$E = \frac{1}{2a}$$

Thus, *x*(*t*) is an energy signal.

# Exercise (1/2)

❑ Determine whether the following signals are energy signals, power signals, or neither. Assume θ=0.

$$X(t)= A\,Cos(w_o t + \theta)$$

**Ans.**

X(t) is periodic

The sinusoidal signal x(t) is periodic with $To = \frac{2\pi}{w_o}$

The power of x(t) is:-

$$p = \frac{1}{T_o}\int_0^{T_o} |x(t)|^2\, dt$$

$$p = \frac{w_o}{2\pi}\int_0^{T_o} |A\,Cos(w_o t)\,|^2\, dt$$

Note: $\cos^2(x)= 0.5\,(1+\cos(2x))$

$$p = \frac{w_o}{2\pi}\int_0^{T_o} A^2\, cos^2(w_o t)\, dt$$

$$p = \frac{A^2 w_o}{2\pi}\int_0^{T_o} \frac{1}{2}[1 + cos(2w_o t)]\, dt$$

$$p = \frac{A^2 w_o}{2\pi}\int_0^{2\pi/w_o} \frac{1}{2}[1 + \cos(2w_o t)]\, dt$$

$$p = \frac{A^2 w_o}{2\pi}\left[\frac{1}{2}\left[t + \frac{1}{2wo} * \sin(2w_o t)\right]\right|_0^{2\pi/w_o}$$

0

$$p = \frac{A^2 w_o}{2\pi} * \frac{1}{2} * \frac{2\pi}{w_o} = \frac{A^2}{2} < 0$$

Note that periodic signals are, in general, power signals.

# Thanks!